

Data Structures Syllabus

NYU — CSCI-UA.0102-060 — Fall 2026

October 4, 2026

Note: the syllabus may be updated during the semester. An up-to-date version will remain available on Brightspace.

Lecture

- **Instructor:** Romain Cosson (rc6142@nyu.edu)
- **Time:** Tuesday / Thursday 3:30 - 4:45 pm
- **Location:** Room 261. 238 Thompson St (GCASL).
- **Office Hours:** Tuesday 4:45 - 5:45 pm, Courant (WWH) Office 401.

Recitation

- **Recitation Leader:** Sameera Bangaru (sb10517@nyu.edu)
- **Time:** Friday 3:30 - 4:45 pm
- **Location:** 31 Washington Pl (Silver Ctr) Room 509.
- **Office Hours:** Friday 2:15 - 3:15 pm (Silver Ctr) Room 510.

Team

- **Course Assistant:** Harshaa Reddy (hg3014@nyu.edu)
- The class is also assigned **one grader**.

Department policies

- **Tutoring:** for all is organized at the department level for CS102: [Visit this link to see the next sessions](#).
- **Prerequisite:** A grade of C or higher at Introduction to Computer Science (CSCI-UA 101) or a successful placement exam. Contact your advisor if you need to switch to a CS101 class.

Topics

The goal of this class is to understand the design of data structures, and their role to obtain efficient and reliable algorithms. Topics include:

- Java and object oriented programming (encapsulation; types; inheritance; polymorphism)
- Abstract data types (lists; stacks; queues; priority queues; dictionaries)
- Implementations (arrays; Linked lists; doubly linked lists; resizable arrays; binary heaps; binary search trees; AVL trees;)
- Algorithmic thinking (sorting; searching; recursion)

Students who are already familiar with the material can test out of the course.

Material

- **Slides:** Used to cover the material for the class (theory).
- **Whiteboard:** Used to go through classic examples and exercises (practice).
- **Code:** Used to implement algorithms seen in class (code).

Prof. Evan Korth and Prof. Joanna Klukowska have graciously provided much of the material used in this class, and they have taught the topic for years.

More material

- **Books:** *Data Structures and Algorithms in Java, 6th edition*, Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser (recommended) and *Data Structures and Algorithm Analysis in Java (third edition)*, Mark Allen Weiss (optional).
- **Coding environment:** All programming work in this course will be done in Java. We recommend using Visual Studio Code (VS Code) rather than Eclipse or other IDEs. Install the Java Development Kit (JDK) from the official Oracle website and verify the installation by running “`java -version`” and “`javac -version`” from a terminal. Configure your workspace so that VS Code recognizes the installed JDK.
- **Brightspace:** Brightspace will be used to post class and recitation content, exercises, grades, and announcements. There will be two distinct Brightspace websites: one for the lecture and one for the recitations.

Assignments

- **Exams:** There will be one midterm and one final. The midterm will take place during class. Time and location will be announced via Brightspace. You should come to me at the beginning of the semester if you have foreseeable constraints (e.g., observed religious holiday) that would prevent you from taking an exam on some date. If you must miss the midterm or final for an emergency reason (e.g., medical), contact me before the start of the exam.
- **Quiz:** Quizzes will consist in a (10-15) minutes evaluation on paper, usually at the end of recitations. There will be 5 to 10 quizzes in the semester.
- **Homework:** Homeworks will mostly consist of programming assignments, posted on Brightspace usually on Fridays (5 to 10 in the semester). Those assignments will be presented and initiated during recitation. You are allowed one late homework submission (4 days max), with no justification required.
- **Project (optional):** The details for an optional project remain to be determined. The project could be used for extra credit.

Questions regarding grading, exams, quizzes, and homework should be asked in class (not by email) so that everyone gets the same information.

Policy for missed quizzes

The students are expected to participate to all quizzes. If you have a valid reason for missing a quiz (e.g., competition, religious holiday, medical appointment), you must send an email to the course assistant at least 1 week before the quiz (unless “*force majeure*”), and in this case, the grade will be dropped. You are not expected to drop more than one grade in the semester (again, unless “*force majeure*”). There will not be make-up quizzes.

Collaboration policy and AI

See the Computer Science Department’s Academic Integrity statement. Oral collaboration between the students is permitted (and encouraged) but code should be your own. The use of AI is allowed to help you understand concepts covered in class, and even to help you if you are stuck with your homework. But your code should not be copy-pasted from a conversation with AI. During the semester, the recitation leader or I may test you by asking you questions about your code. You are expected to be able to explain every line of code that you submit for evaluation.

Final grade

The final grade (G) will be a weighted combination of final exam (E) midterm (M) quizzes (Q) and homeworks (H). The **tentative** weights are:

$$G = 0.6 \times (0.6E + 0.4M) + 0.3Q + 0.1H.$$

Conclusion

Your ideas to promote a better understanding and learning in class are welcome. Feel free to ask questions, to engage in discussions.